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A Call for a Unified Database to Address Exposure Disparities in the United States

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Abstract

United States (US) drinking water quality is federally regulated under the Safe Drinking Water Act (SDWA) with the goal of ensuring clean, safe drinking water for the entire population. However, siting of pollution point sources in historically redlined communities with lower socioeconomic status and higher proportions of people of color adversely affects drinking water quality. Prior studies show higher concentrations of a suite of contaminants in small PWS, PWS in rural areas, and PWS serving greater proportions of people of color. Drinking water crises in Flint, Michigan (ca. 2014) and Jackson, Mississippi (ca. 2022) are visible examples of a chronic and widespread pattern of elevated exposures in certain communities. Comprehensive state monitoring data produced as part of compliance with the SDWA are presently not accessible to the public. Further, metadata on water treatment technology and sociodemographic composition of customers served are lacking. Here we argue that improving data transparency is essential for spearheading action to address exposure disparities. We propose the creation of a unified national public database on water quality, treatment technology, and customer metadata with accompanying visualizations for the public that leverage new AI tools. Public action to address exposure disparities gained momentum over the last 30 years but was halted abruptly in 2025. Therefore, new partnerships are urgently needed to continue the pursuit of environmental justice.

Graphical Abstract

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CONFLICTS OF INTEREST
The authors declare no conflicts of interest for this article.

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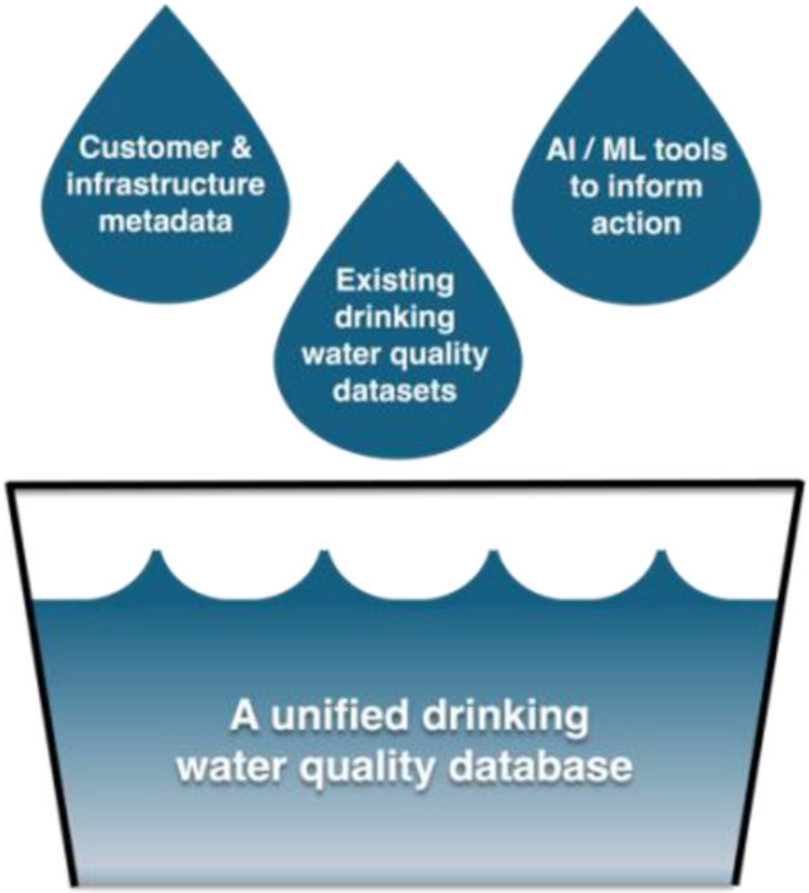

INTRODUCTION

Goal 6 of the United Nations Sustainable Development Goals calls for "safe and affordable drinking water for all" by 2030 (United Nations General Assembly, 2015, p. 6). Drinking water quality has been regulated federally in the United States (US) since 1974 as part of the Safe Drinking Water Act. Yet, more than 130 million US residents (ca. 2016–2019) rely on drinking water that violates maximum contaminant levels (MCLs) (Kennelly, 2022). Adverse health implications have been associated with exposure to regulated (Spaur et al., 2024) and unregulated drinking water contaminants (Luo et al., 2025; Waterfield et al., 2020). Additionally, only 90 contaminants have been regulated in drinking water (US EPA, 2015c) among the over 350,000 chemicals that were registered for production and use (Wang et al., 2020). Up to 200 million US residents consume water containing detectable per- and polyfluoroalkyl substances (PFAS), an important new class of drinking water contaminants (Andrews & Naidenko, 2020; Tokranov et al., 2024). The MCLs for 6 PFAS in 2024 were the latest Federal drinking water standards to be finalized for new contaminants in several decades (US EPA, 2024a). However, these were partially rescinded in 2025, emphasizing the slow rate of regulatory changes (US EPA, 2025).

Access to drinking water in the US is shaped by the diverse range of public and private systems serving the population, each with varying levels of oversight and regulation. There are nearly 150,000 public water systems (PWS) across the US that range in size from very small (serving 25 to 500 customers) to very large (serving greater than 100,000 customers) (US EPA, 2024d; US EPA Office of Water, 2015b). An additional 43 million individuals obtain their drinking water from private wells, which are not subject to routine monitoring as part of Federal regulations (USGS, 2019).

Elevated exposures to drinking water contaminants are known to be more prevalent in historically disadvantaged communities across the US (Dai et al., 2025; Kennelly, 2022). Public health emergencies like those in Flint, Michigan ca. 2014 and Jackson, Mississippi ca. 2022 (Oyshi et al., 2024; Pauli, 2020) raised widespread awareness of drinking water exposure disparities. However, these are only the most visible examples of a much more chronic and widespread pattern of elevated exposures in certain communities. For example, higher concentrations of pollutants like arsenic, nitrates, and others have been most frequently reported in marginalized communities, including communities with high proportions of people of color, non-English speaking households, and lower socioeconomic status (Dai et al., 2025; Nigra et al., 2020; Schaider et al., 2019). Past work has noted disparities in both the siting of PFAS sources and detection in PWS serving greater proportions of people of color (Liddie et al., 2023; Maruzzo et al., 2025; Mueller et al., 2024) as well as socioeconomic disparities in rural areas (Liddie et al., 2023).

Regulatory actions to address environmental health inequalities in the United States had been gaining momentum over the past three decades. A 1994 Executive Order called for action to address disparities that pose an unfair burden on minority and low-income populations (US EPA, 2013). In 2024, the Biden administration provided support for investments in drinking water treatment technology under the Bipartisan Infrastructure Law to address US communities disproportionately exposed to contaminants (The White House, 2021a). Support for these programs was bolstered by the concurrent Justice40 Initiative, which directed at least 40% of Federal support for critical clean water infrastructure to disadvantaged and disinvested communities (The White House, 2021b). However, Federal support for actions addressing environmental disparities was abruptly halted in January 2025 (The White House, 2025b), creating a critical need for alternate strategies to protect the health of vulnerable communities.

In this *Perspective*, we argue that urgent action is needed to address drinking water quality disparities across the US. To this end, we propose three steps that the environmental justice research community—in partnership with other researchers, communities, governmental and non-governmental organizations, and industry—could undertake to address disparities in drinking water contaminant exposures. Together, these steps would not only help advance our understanding of the health effects of drinking water exposures and prevent harms by identifying and addressing unsafe water, but also build public trust in drinking water where it is safe and reduce reliance on costly or unhealthy alternatives such as bottled water or sugar-sweetened beverages. These steps include the following:

A. **Improve transparency and availability of drinking water monitoring data** for regulated and unregulated contaminants in both PWS and private wells.

B. **Develop new tools and databases to identify underserved communities and gaps in existing treatment infrastructure** through new collaborations and partnerships with state monitoring programs and the public.

C. **Leverage the ongoing revolution in artificial intelligence (AI) and machine learning (ML) to improve drinking water quality** by bolstering the visualization and dissemination of these data to the public in ways that empower communities to take informed action.

For each of these recommendations, we outline the *status quo* and associated challenges, meaningful steps toward these goals that could be taken now, and "stretch goals" (future priorities) that would rapidly accelerate our progress toward these aims.

STEPS TOWARD A UNIFIED DRINKING WATER QUALITY DATABASE

1. Improve transparency and availability of drinking water monitoring data

Status quo: The Safe Drinking Water Information System (SDWIS) is maintained by the US EPA and provides information on drinking water violations for regulated contaminants for PWS (US EPA, 2015b). In total, more than 300 million Americans rely on PWS (Levin et al., 2023). Violations are reported quarterly and are split into monitoring/reporting and health violations. Health violations are based on whether a system is above an MCL as specified in the SDWA (US EPA, 2015b). SDWIS does not consistently include continuous concentration data, especially below established MCLs. As a result, most prior studies of national water quality have modeled categorical or binary drinking water quality indicators rather than continuous concentration (US EPA, 2015b). This is summarized in our literature review detailed in the Supporting Information (see SI and Figure 1 for a visual summary of studies found). For example, binary drinking water quality indicators could refer to whether concentrations are above or below cutoff values (e.g., a method detection limit [MDL] or MCL). This presents a problem because these binary indicators necessarily result in the loss of information about the continuous range of true exposures. Further, inconsistent reporting among primacy agencies due to differing state drinking water guidance and/or lack of federal MDL standards may misconstrue PWS occurrence data for some contaminants of human health concern. Regulatory thresholds tend to shift toward lower values (e.g., arsenic) over time as health studies advance understanding of effects at lower exposure levels (US EPA, 2000). MCLs themselves often consider technical feasibility as well as economic impacts rather than just potential drinking water contaminant exposure risks, potentially setting thresholds above those that are most protective of public health (Bradley et al., 2018).

The federal government presently manages the Unregulated Contaminant Monitoring Rule (UCMR) that monitors selected unregulated contaminants in PWS every five years. Historically, all large PWS (serving >10,000) and a representative group of smaller PWS (serving ≤10,000) have participated in these recurring monitoring programs (US EPA, 2024c). Although large PWS serve most of the US population that receives public water (approx. 82%), smaller systems serving fewer than 10,000 people comprise the largest number of PWS (approx. 92%) (US EPA Office of Water, 2011). Prior monitoring requirements for UCMR have resulted in the underrepresentation of these smaller systems across UCMR cycles (Figure 2) and systems serving Tribal Nations (Mok et al., 2022). To bridge some of this gap, capacity for all PWS serving 3,300–10,000 people to participate in the ongoing round of UCMR testing (UCMR 5) has been proposed for newer UCMR cycles, alongside a representative sample of smaller PWS (US EPA, 2024c). However, current estimates indicate that UCMR 5 may still underrepresent populations served by Tribal PWS in comparison to non-Tribal PWS. These reflect differences in population sizes served by non-Tribal compared to Tribal PWS, which include a larger proportion of systems serving ≤3,300 people in number and as a proportion of their service populations (Mok et al., 2022).

The Environmental Working Group's Tap Water Database (Environmental Working Group, 2021) has synthesized state and Federal data on drinking water contaminants with support from private funding. *Ad hoc* efforts synthesizing state data have also been conducted by academic groups and a diverse group of statewide monitoring databases (Hu et al., 2023; Liddie, 2023). In many cases, these efforts are a combination of publicly available sources, requested data, and data obtained via Freedom of Information Act requests. While these data have been extremely valuable for understanding patterns of drinking water contamination across the country, a more formal and systematic approach to data synthesis and sharing would reduce redundant efforts across groups.

Monitoring data for the 43 million Americans who rely on private wells are more poorly characterized compared to monitoring data for individuals served by PWS (USGS, 2019). Private wells are unregulated drinking water sources defined as serving fewer than 25 people or 15 service connections for at least 60 days during the year (US EPA, 2015a). Beyond their unregulated status, additional challenges due to costs to individuals and conflicting information have impeded the effectiveness of private well outreach programs (Morris et al., 2015). Currently, initiatives such as the United States Geological Survey's National Water-Quality Assessment project (DeSimone, 2009) and state-level programs (e.g., PFAS sampling conducted in New Hampshire for private wells (New Hampshire Department of Environmental Services, 2025) and in various other states, compiled in Table S2) have conducted and/or compiled private well monitoring data. However, to our knowledge, no systematic, publicly available dataset of private well monitoring exists for research purposes or visualization.

Immediate priorities: A lower-hanging goal is to compile existing databases on drinking water quality into a unified database that abides by Finding, Accessibility, Interoperability, and Reusability (FAIR) principles. These principles, which are now priorities in all federally funded research, seek to improve data accessibility and reusability (Wilkinson et al., 2016). Data in an initial version of this database would include: (a) longitudinal data for PWS on violations including monitoring, reporting, and health-based violations obtained from SDWIS (US EPA, 2015b); (b) concentrations of regulated contaminants (e.g., obtained from statewide data and US EPA's Six-Year Reviews (Environmental Working Group, 2021; US EPA Office of Water, 2014); and (c) concentrations of unregulated contaminants from available statewide and federal monitoring (e.g., the PFAS statewide sampling dataset, other statewide monitoring, and federal UCMR cycles (Liddie, 2023; US EPA, 2024c)). Ideally, drinking water sampling data would be accompanied by analytical methods and either MDL or reporting limits, which are important for statistical interpretation.

Coordination among academic and non-profit organizations with experience compiling federal and/or statewide data is needed to complete this initial effort. Key initial challenges in maintaining these data include continuous tracking of statewide and federal monitoring as they become available; inclusion of accompanying QA/QC data and an understanding of how these data may be reported differently from state to state or over time; and taking appropriate measures to standardize monitoring data from several sources. An initial end-goal for this compilation effort would comprise a unified database that can be queried by individuals and community organizers to learn about local drinking water quality to make informed decisions about potential exposures and address related environmental justice concerns. Larger scale queries, such as those available currently through SDWIS (US EPA, 2015b), would also enable researchers to more seamlessly conduct exposure and environmental justice research on drinking water quality. Ultimately, we recommend that a non-governmental institution host this unified database due to the politicized nature of drinking water quality and environmental justice concerns in the US.

Future priorities: In an ideal world, a unified national database on drinking water quality would also include private well monitoring data. Data would need to be deidentified and/or spatially aggregated to protect the privacy of homeowners. Any advancement in this area would represent a major improvement over existing databases that primarily include PWS. This could include monitoring obtained from statewide private well monitoring programs and private well water quality testing conducted by occupants or laboratory services, which both represent major untapped data sources. Innovative efforts to collect and homogenize such data through efforts such as a smartphone app to crowdsource data from private well owners could be explored. Such data could be flagged with identifiers that distinguish them from systemic monitoring efforts by state and federal agencies. Key challenges in compiling data obtained from crowdsourcing include data duplication, privacy concerns, lack of accompanying QA/QC data, and non-standard testing methods. For this reason, future data quality assessment procedures for these data might include ongoing comparisons of self-reported monitoring results with pre-existing statewide monitoring or accredited laboratory-reported results for a smaller, geographically diverse subset of private wells. As these data are compiled more systematically, they can be used to identify disparities in monitoring or maintenance, which have been identified as issues for both private wells (Hayes et al., 2025) and PWS (Glade & Ray, 2022). This, in turn, would help identify chronically underserved areas that warrant additional attention and outreach, following examples that used low-cost air sensor networks to examine air quality (Mousavi et al., 2021). In summary, a combination of crowdsourced data on private wells and partnerships with states to incorporate historic and ongoing private well surveillance efforts and mandates (Table S2) could dramatically improve present understanding of exposures and risks associated with drinking water contaminants across the US.

2. Develop new tools and databases to identify underserved communities and gaps in existing treatment infrastructure

Status quo: Our literature review (see Supporting Information and Table S1) revealed that studies on drinking water quality disparities often analyze data at spatial scales different from their native resolution. Drinking water quality monitoring data is often combined with US census data (e.g., by county, county subdivision, or zip code tabulation area) due to a historical lack of sociodemographic information available at the native PWS service area resolution (VanDerslice, 2011). This creates a modifiable areal unit problem as the choice of spatial unit can change results and their interpretation when data are aggregated (Baden et al., 2007; Dai et al., 2025). For example, multiple PWS and areas relying on private wells may be included within the same census boundary. Dai et al. (2025) showed that associations with sociodemographic data based on county-level aggregation indicated generally higher effect sizes with wider variances and fewer statistically significant associations than associations at the native PWS service area scale.

In 2024, a nearly comprehensive dataset of community water system service area boundaries (covering 99% of the population served by these systems) was released by the US EPA (US EPA Office of Water, 2024b). Community water systems are PWS that serve the same population year-round. This data release is a follow-up to earlier data products, including SimpleLab's Tiered, Explicit, Match, and Model dataset, the US Geological Survey's Public-Supply Water Service Areas datasets, and the Municipal Drinking Water Database (Buchwald et al., 2024; Hughes et al., 2023; SimpleLab, EPIC, 2022). The first release of US EPA's dataset relies largely on modeled boundary data, with estimated boundaries for at least 28 (60%) states (US EPA Office of Water, 2024b). This improved, federally maintained, publicly available nationwide database helped in identifying drinking water challenges through the (formerly available) US EPA's environmental justice screening and mapping tool (US EPA, 2024b) (Figure 3), which is currently being archived (The Public Environmental Data Partners, 2025).

Systemic disparities in drinking water quality are deeply tied to the technical, managerial, and financial capacities of public water systems and lead to many water systems failing to meet federal standards (Balazs & Ray, 2014). Utility staffing and maintenance are known issues that contribute to environmental injustice, especially for small PWS serving low-income communities (Glade & Ray, 2022). Unfortunately, data related to these factors are mostly unavailable at a federal level, leaving researchers and policymakers without the tools necessary to fully understand and address these disparities. It is therefore unsurprising that most studies identified in our literature review (see Table S1 and a prior review by Vanderslice, 2011) fail to consider the underlying factors contributing to drinking water quality disparities.

PWS are responsible for the operation and maintenance of their own distribution systems, while property owners are responsible for maintenance of piping within their properties and (if applicable) their private water infrastructure (US EPA Office of Water, 2020b). Detailed data on distribution systems may be key data sources to assess environmental justice issues with respect to copper and lead contamination. Some cities and states (including

Massachusetts; Michigan; New York City, New York; Toledo, Ohio; Trenton, New Jersey; and Washington, District of Columbia) have created public digital inventories of lead service lines (City of Toledo, 2024; DC Water, 2019; Massachusetts Department of Environmental Protection, 2024; MI Department of Environment, Great Lakes, and Energy, 2024; New York City Department of Environmental Protection, 2021; Trenton Water Works, 2024; Webb et al., 2017). As of October 2024, all stage primacy agencies were required to prepare and maintain service line inventories of pipe material classifications with location identifiers for lead pipes (US EPA Office of Water, 2023). These inventories have helped identify new areas in the US that have a history of lead service line installations but were previously unknown.

Immediate priorities: Combining water quality data with service area boundaries, such as those available in the US EPA's Community Water System Boundary dataset (US EPA Office of Water, 2024b), into a visualizable and query-ready tool is a clear first step for increasing data accessibility. An important addition to these data would be PWS-level sociodemographic information, which are not directly surveyed or assessed by PWS. However, PWS-level sociodemographic composition can be derived by combining raster-based layers on readily available factors such as population density (Facebook Connectivity Lab and Center for International Earth Science Information Network (CIESIN) - Columbia University, 2020), social vulnerability layers (e.g., the Socioeconomic Data and Applications Center (Columbia Climate School Center for Integrated Earth System Information, 2025), which has plans to transition to alternative locations for storage), and census data. These geospatial approaches are growing in recent studies (Dai et al., 2025; Statman-Weil et al., 2020) and should continue in parallel as the data quality of US EPA's service area boundary dataset continues to improve.

Another important dimension of the proposed unified national database is including information on treatment processes and infrastructure. Some current and historical data have been archived in SDWIS (US EPA, 2015b) and are extremely useful for understanding and interpreting PWS drinking water contaminant concentrations and violations. In our past work, we found that not all historical data are publicly available. Older historical data (>10 years) can be obtained via Freedom of Information Act requests (Liddie, 2025). Findings from research related to these data could fill an important gap, as the operational, physical, and regulatory aspects of water infrastructure are rarely directly investigated (VanDerslice, 2011). Lowering barriers to public accessibility to these data could help community groups formulate actions to improve drinking water quality at the local scale and inform individuals concerned about household exposures.

Future priorities: A key future priority should be resolving utility-level information on service area boundaries and customer sociodemographic composition for all PWS. Making such data available requires coordination with states and utilities to continue to generate publicly available service area boundary maps (US EPA Office of Water, 2024b). Doing so would reduce the reliance on modeled or estimated boundaries, which may introduce error both in relation to individual-level exposure assessment (and queries) and analyses of sociodemographic disparities in contamination.

Augmenting currently available SDWIS treatment process data with information related to target contaminants, removal efficiencies, staffing, and funding would be a laborious but valuable future priority for inclusion into a unified drinking water quality database. Integrating existing distribution system datasets, such as the installation of lead or copper service lines, would be another invaluable addition to the proposed database. The Biden administration set a 10-year timeline to replace all lead service lines and the Bipartisan Infrastructure Law dedicated $15 billion specifically for lead pipe replacement, with nearly half of these funds formerly directed to disadvantaged communities (The White House, 2024). However, future disbursement of funds is presently uncertain (The White House, 2025a). Compiling progress on lead service line replacement within a visualization tool could help inform actions to continue replacements, even in the absence of Federal support. Existing maps, such as the Environmental Defense Fund's map of lead service line replacement programs and Grist's map of infrastructure-related projects under the Biden administration, offer an important first step in identifying where lead service line replacements have been enacted or are ongoing across the US (Aldern, 2025; Environmental Defense Fund, 2024). However, additional engagement with cities and utilities may be needed to disaggregate these replacements at a finer geographic scale (as in the aforementioned state inventories).

3. Leverage the ongoing artificial intelligence (AI) and machine learning (ML) revolution to improve access to drinking water quality data

Status quo: Machine learning (ML) algorithms are increasingly being used to interpret large environmental datasets, including those on drinking water quality (Lombard et al., 2021, 2024; Tokranov et al., 2024). Some technical barriers to easy implementation of these algorithms remain, as discussed elsewhere (Hu et al., 2023). Past drinking water studies have typically focused on predicting binary outcomes (e.g., concentrations above a threshold), with more variable prediction metrics reported for multiclass outcomes and continuous concentrations (Hu et al., 2023).

These modeling methods show great promise for assisting with data visualization in a publicly accessible way and creation of exposure surfaces that can be linked to epidemiologic studies (e.g., (Luo et al., 2025) and (Spaur et al., 2024)). Such approaches follow numerous studies in the air pollution epidemiology literature, in which prediction modeling approaches have frequently been employed to conduct spatiotemporal exposure assessments for large, nationwide cohorts (e.g., (Di et al., 2016, 2017; Ma et al., 2022)). Machine learning approaches are also uniquely poised to help overcome gaps in private well monitoring for specific contaminants (e.g., (Hu et al., 2021) and (Lombard et al., 2021) at the state and national scales, respectively). These more immediate priorities would be well-supported by the curation of a unified drinking water quality database proposed here of regulated and unregulated contaminants.

Immediate priorities: The ongoing revolution in artificial intelligence (AI) combined with advances in computation and storage continues to facilitate methods for interpreting, visualizing, and organizing data in a manner that can interactively inform our needs. We identified several existing drinking water quality visualization tools at the state and national scale, although these tools typically focus on single groups of contaminants or types of data (e.g., drinking water quality data alone, but not water system characteristics or information about the communities they serve) (Table S3). A more comprehensive nationwide visualization tool would therefore greatly improve access to these data and may more readily shed light on the real communities in which regular monitoring is lacking. Simple prompt-based data visualizations (including maps of measured results and predicted exposure surfaces) would also improve capacity for designing monitoring programs that target both the most vulnerable communities and the most impacted watersheds by various contaminants. Other templates for successful sharing include Wikipedia as a living example of a common information repository and open-source mapping tools like openlayers.org for mapping and visualizing information.

Future priorities: Collaborative opportunities for AI-assisted tools on drinking water quality have the potential to empower the public and researchers concerned about these issues. We envision a future in which AI-assisted generation of data visualization and interpretation will facilitate access to drinking water quality data. This priority is not without challenges, such as an expert-vetted understanding of the limits of AI tools in accurately completing tasks related to drinking water topics (Zhu et al., 2023) and environmental justice issues. Integration of techniques like retrieval-augmented generation would help reduce inaccurate responses and provide evidence-based recommendations to users informed by local data.

AI tools, such as large language models, may also hold promise for automating data processing steps, particularly when harmonizing disparate data formats. For example, researchers have proposed that large language models can simplify data handling when harmonizing disparate data sources, such as electronic healthcare databases (as described in Painter *et al.*, 2025). This approach is unique in that it has promise to harmonize disparate datasets without requiring a single standardized output format, preserve details across disparate data sources, and generate code to conduct queries. For unifying electronic healthcare data, Painter *et al.*, 2025 stress that these data processing steps are conducted responsibly with the necessary privacy and governance structures. With respect to drinking water quality data, additional work is needed to determine how privacy and governance should be handled when combining disparate data sources, particularly when introducing crowdsourced data or other data sources that have traditionally not been publicly available.

New collaborations between environmental (health) scientists, data scientists, and computer scientists are thus needed to facilitate this transition. Computer scientists are increasingly outlining an "AI for Social Impact" agenda (Tambe et al., 2025). This agenda places positive societal impacts, particularly for underserved groups, as the primary objective of AI research and development. Key elements of this agenda include engaging intended users within the design process and encouraging joint evaluation that allows researchers and designers to learn the limits of AI to accomplish particular tasks (Tambe et al., 2025). This is particularly relevant for the long-term utility and effectiveness of the proposed visualization tools.
In all, the integration of AI-assisted tools with unified drinking water quality data has important implications for environmental justice due to its potential to democratize technical information for general audiences.

CONCLUSION

Addressing drinking water quality disparities in the US requires urgent action and collaboration among researchers, states, non-profit groups, and community groups. Our overarching recommendation is to create a first-of-its-kind unified drinking water quality database, combining the efforts of existing datasets, utility-level service area boundaries, customer sociodemographic composition, infrastructural data, and tools to support community involvement. A unified database could also act as an exposure assessment tool to understand the health effects of drinking water exposures and benefits of reduced exposures, which have been chronically underexamined (Brown et al., 2023; Villanueva et al., 2014).

We present Table 1 with an overview of the immediate and future priorities to create this database, as well as currently anticipated challenges. Although many of the extant, disparate datasets are currently free to access, funding is needed to enable researchers to compile, store, and continuously update these data. This requires additional work on the part of researchers to identify and convene stakeholders across various sectors (e.g., academic, nonprofit, industry, governmental, and non-governmental organizations) to establish long-term funding mechanisms. Additional specific steps for EJ researchers to support this work and better ensure its long-term sustainability and utility for the public include:

1. An illustration of use cases that demonstrate the benefits of a unified national drinking water quality database through their existing community partnerships.
2. Funding opportunities to support methods and data infrastructure development, community-centered data collection, student involvement, and cross-sectoral collaboration.
3. The development of common data standards, metadata structures, and principles for data harmonization for their own ongoing studies.
4. Training for community members and citizen scientists on data collection, analysis, and interpretation of drinking water quality data and visualizations.

In summary, a unified US drinking water quality database managed by non-federal entities would ensure the longevity of accompanying data. AI-assisted tools have the potential to democratize access and interpretation of these curated data. With them, the advancement of environmental justice would depend less on the volatility of politics at the federal level.

Supplementary Material

Refer to Web version on PubMed Central for supplementary material.

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DATA AVAILABILITY

The data used to create the figures in this article were derived from the following resources available in the public domain: the Safe Drinking Water Information System (US EPA, 2015b) and the Third, Fourth, and Fifth Unregulated Contaminant Monitoring Rules (US EPA Office of Water, 2015a, 2020a, 2024a).

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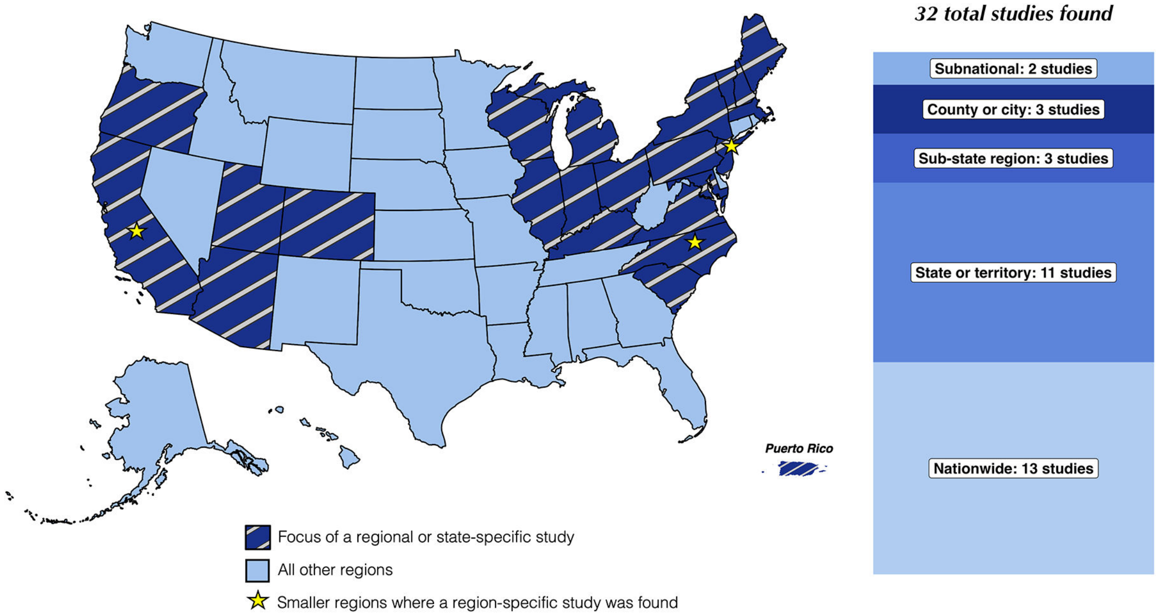

Figure 1: Summary of empirical studies on U.S. drinking water quality disparities

Note: States and territories are not shown to scale. For ease of visibility, Alaska, Hawaii, and Puerto Rico are not shown in their true locations. Table S1 contains more information on each study and a brief description of the primary findings. Studies that conducted both nationwide and sub-national analyses are categorized as nationwide in this figure.

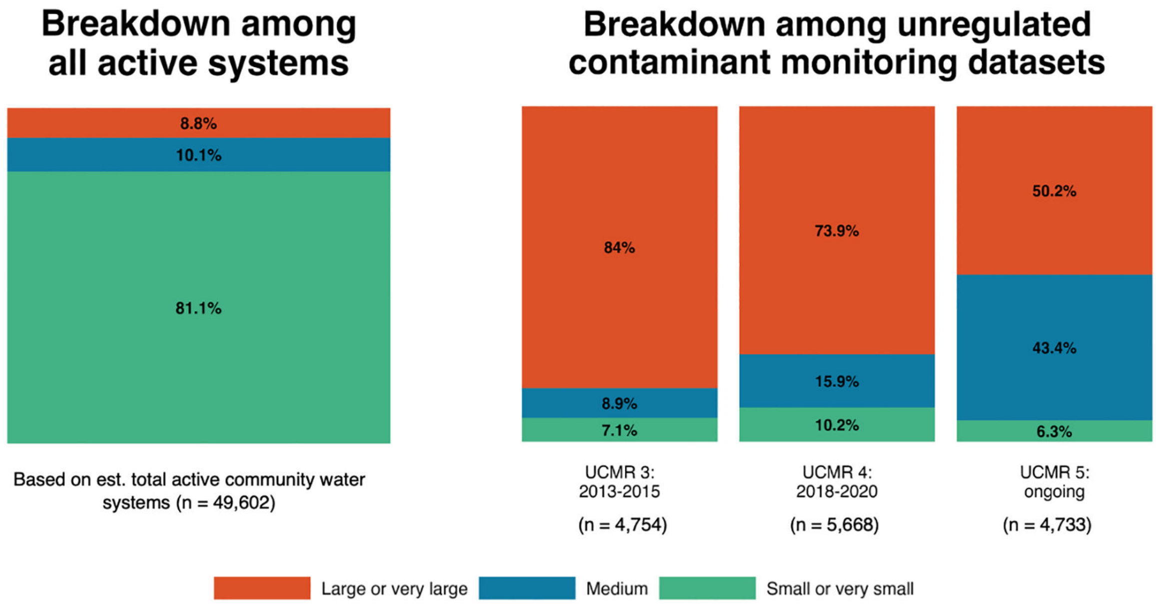

Figure 2: Breakdown of the number of community water systems nationwide and within recent Unregulated Contaminant Monitoring Rule (UCMR) datasets

Note: Community water systems are public water systems that serve the same population year-round. Total active community water systems were enumerated using the Safe Drinking Water Information System (US EPA, 2015b). Data from the third, fourth, and fifth cycles of UCMR are enumerated on the right (US EPA Office of Water, 2015a, 2020a, 2024a).

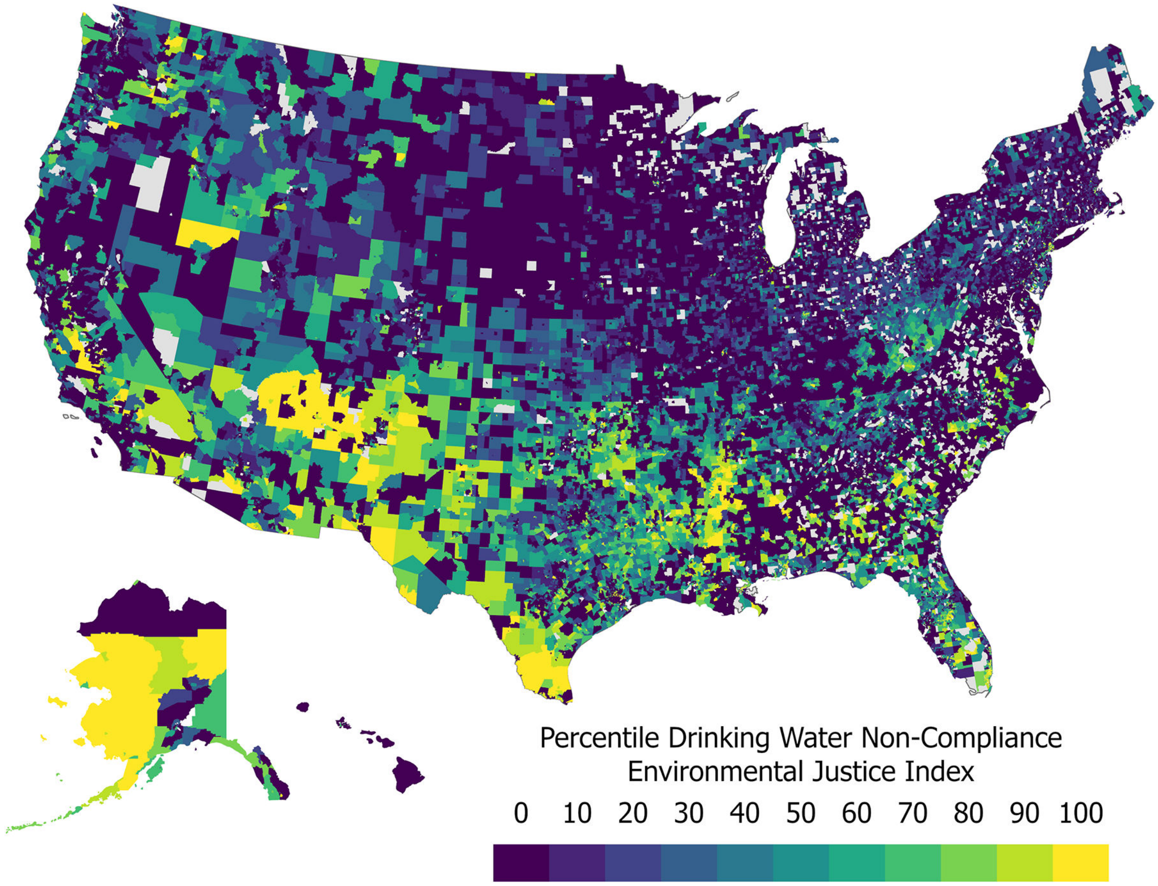

Figure 3: Map of the US EPA drinking water non-compliance environmental justice index (as percentiles)

Note: This map shows percentiles of the drinking water non-compliance environmental justice index from the US EPA's Environmental Justice Screening and Mapping Tool (US EPA, 2024b). Scores reflect overlapping Safe Drinking Water Act violations by community water systems and populations of low-income or people of color, where high scores indicate higher environmental burdens.

Table 1:

Steps toward a unified drinking water quality database

Step	Immediate priorities	Immediate challenges	Future priorities	Future challenges
1. Improve transparency and availability of drinking water monitoring data	Combine existing datasets on regulated and unregulated drinking water contaminants in community water systems	Disparate datasets with different reporting formats	Incorporate crowdsourced deidentified or aggregated data on drinking water quality for private wells	Crowdsourced data are accompanied by data quality and privacy concerns
		Funding and hosting constraints		Long-term, multisectoral partnerships and funding needed
2. Develop new tools and databases to identify underserved communities and existing treatment infrastructure	Incorporate resolved, utility-level data on sociodemographic composition	Service area boundaries maps are not yet available for some states	Incorporate geospatial data on complete, national service area boundary maps	Many of these factors are not yet available nationwide
	Incorporate existing treatment process information in an easily accessible format	Ongoing engagement with multiple stakeholders (e.g., academics, non-profit groups, and states) should be established	Incorporate additional data related to removal efficiencies, staffing, funding, and distribution systems	
3. Leverage the ongoing revolution in artificial intelligence (AI) and machine learning (ML) to improve access to drinking water quality data	Open data repositories and visualization tools	Curation of data sources above is necessary	Use of AI techniques to streamline data processing	Vetted understanding of security and governance when combining, analyzing, and visualizing disparate datasets
			Incorporation of AI-assisted tools to inform and empower communities	Vetted understanding of the limits of these tools when presented drinking water quality prompts and their interpretation of environmental justice issues